

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$([g,h])^\circ = (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$$

the converse of a fold over decimals undoes the algebra $[g,h]$ and then either **wrap**, or the fold's converse on the front and **snoc**

1	$[g,h]^\circ$	$(\emptyset) = [\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}(\emptyset)) \emptyset$
2	$= ([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}([g,h])^\circ)) [g,h]^\circ$	$(R + S) [P,Q] = [R P, S Q]$
3	$= ([\text{wrap}, \text{snoc}]^\circ [g, \neg \text{Digit}([g,h])^\circ] h)^\circ$	$[U,V]^\circ [R,S] = (U^\circ R) \cup (V^\circ S)$
4	$= ((\text{wrap}^\circ g) \cup (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ) h)^\circ$	$(R \cup S)^\circ = S^\circ \cup R^\circ$
5	$= (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ) h^\circ \cup (\text{wrap}^\circ g)^\circ$	$R \cup S = S \cup R$
6	$= (\text{wrap}^\circ g)^\circ \cup (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ) h^\circ$	$(R S)^\circ = S^\circ R^\circ$
7	$= (g^\circ \text{ wrap}) \cup ((\neg \text{Digit}([g,h])^\circ) h^\circ \text{ snoc})$	$(R S)^\circ = S^\circ R^\circ$
8	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$	$\neg \text{Digit}(R)^\circ = \neg \text{Digit}(R^\circ)$
9	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$	

$\text{val}^\circ = (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{ snoc})$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

1	$\text{op}^\circ(m,p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$ op° splits a number into its quotient and remainder by 10
2	$(\exists p. 0 < \pi_1(p) \wedge \text{op}^\circ(m,p)) \iff 10 \leq m$ op° gives a pair with a nonzero first component exactly at the numbers with two or more digits
3	$(\exists d. \text{embed}^\circ(m,d)) \iff 0 < m \wedge m < 10$ embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

$\text{Monotonic}(\varphi) \triangleq \forall X. \forall Y. X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$ — φ maps the hom-set $A \rightarrow B$ into itself with $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ is the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

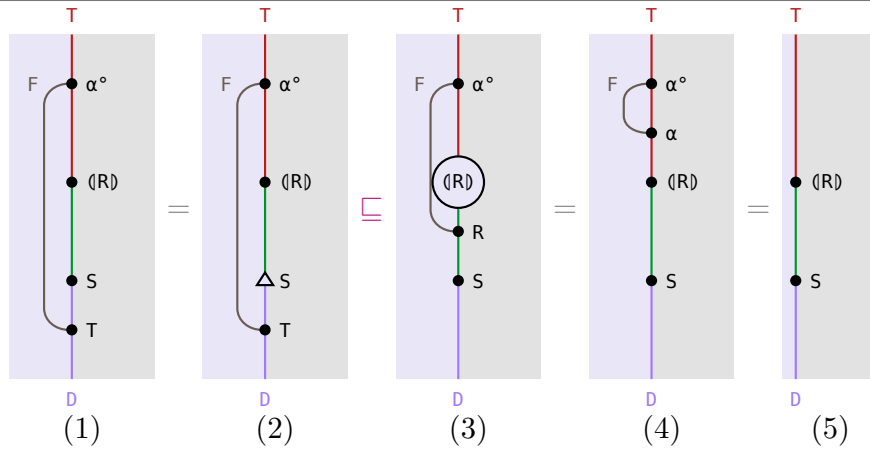
(6.2b)

the law	what it says
1 $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
2 $\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski: the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
3 $\alpha^\circ F(X) R \sqsubseteq X \implies \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
4 $X \sqsubseteq \alpha^\circ F(X) R \implies X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

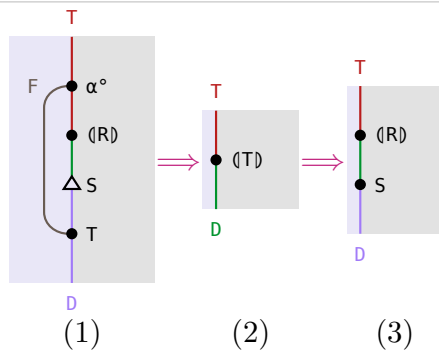
$$F(S) \ T \sqsubseteq R \ S \implies \langle T \rangle \sqsubseteq \langle R \rangle \ S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



1 $\alpha^\circ F(\langle R \rangle S) T$	
2 $= \alpha^\circ F(\langle R \rangle) F(S) T$	$F(S) \ T \sqsubseteq R \ S$
3 $\sqsubseteq \alpha^\circ F(\langle R \rangle) R \ S$	$\alpha \langle R \rangle = F(\langle R \rangle) R$
4 $= \alpha^\circ \alpha \langle R \rangle S$	$\alpha^\circ \alpha = \mathbb{1}$
5 $= \langle R \rangle S$	

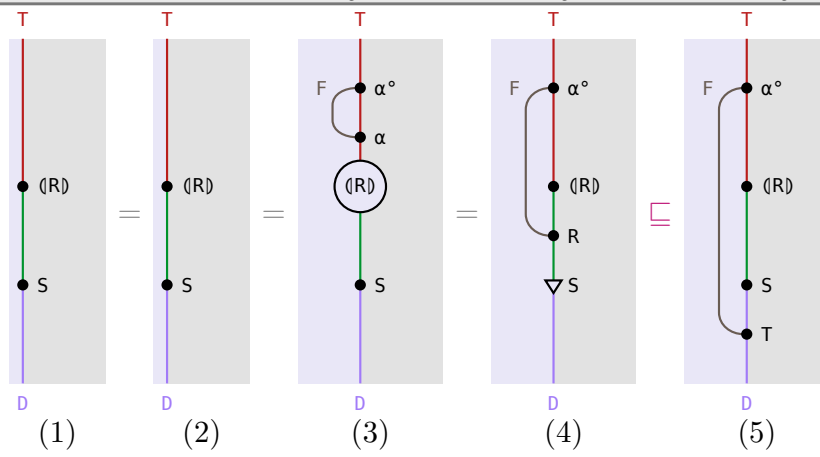


1 $F(S) \ T \sqsubseteq R \ S$	$F(S) \ T \sqsubseteq R \ S$ $\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S$
2 $\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S$	$\alpha^\circ F(X) R \sqsubseteq X$ $\implies \langle R \rangle \sqsubseteq X$
3 $\implies \langle T \rangle \sqsubseteq \langle R \rangle S$	

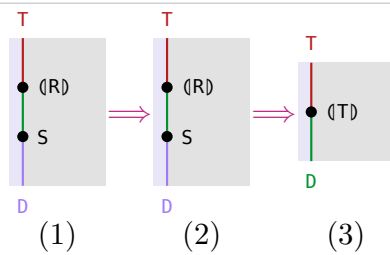
$$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



1	$\langle R \rangle \ S$	
2	$= \langle R \rangle \ S$	$\alpha^\circ \ \alpha = 1$
3	$= \alpha^\circ \ \alpha \ \langle R \rangle \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
4	$= \alpha^\circ \ F(\langle R \rangle) \ R \ S$	$R \ S \sqsubseteq F(S) \ T$
5	$\sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	



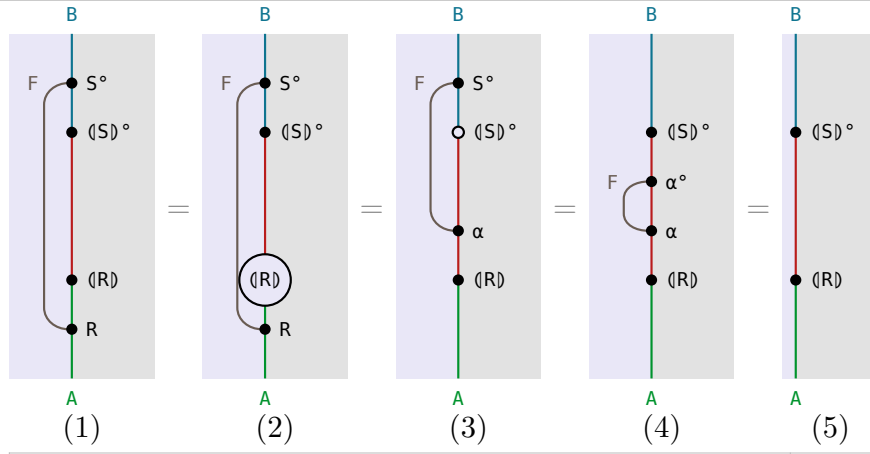
1	$R \ S \sqsubseteq F(S) \ T$	$R \ S \sqsubseteq F(S) \ T$ $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$
2	$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	$X \sqsubseteq \alpha^\circ \ F(X) \ R$ $\Rightarrow X \sqsubseteq \langle R \rangle$
3	$\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$	

§6.3. Hylomorphisms

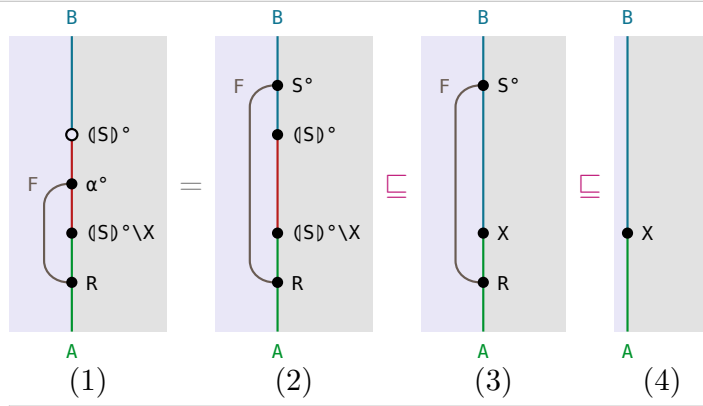
$$(S)^\circ (R) = (\mu X : S^\circ F(X) R)$$

(6.3a)

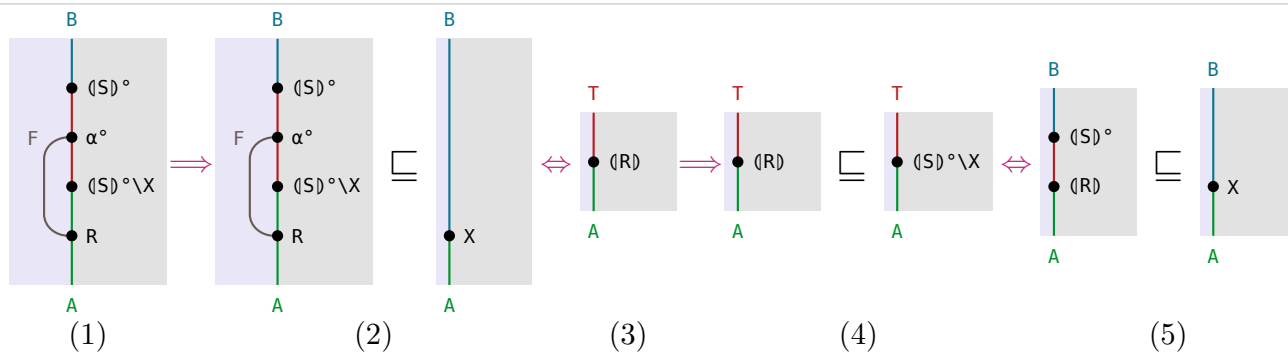
hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



1	$S^\circ F((S)^\circ (R)) R$	
2	$= S^\circ F((S)^\circ) F((R)) R$	$\alpha (R) = F((R)) R$
3	$= S^\circ F((S)^\circ) \alpha (R)$	$(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$
4	$= (S)^\circ \alpha^\circ \alpha (R)$	$\alpha^\circ \alpha = 1$
5	$= (S)^\circ (R)$	



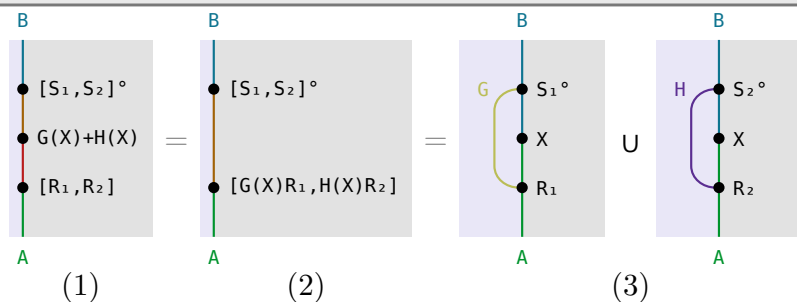
1	$(S)^\circ \alpha^\circ F((S)^\circ \setminus X) R$	$(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$
2	$= S^\circ F((S)^\circ ((S)^\circ \setminus X)) R$	$S (S \setminus R) \subseteq R$
3	$\subseteq S^\circ F(X) R$	$S^\circ F(X) R \subseteq X$
4	$\subseteq X$	



1	$S^\circ F(X) R \subseteq X$	$S^\circ F(X) R \subseteq X$
2	$\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$	$\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$
3	$\Leftrightarrow \alpha^\circ F((S)^\circ \setminus X) R \subseteq (S)^\circ \setminus X$	$X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$
4	$\Rightarrow (R) \subseteq (S)^\circ \setminus X$	$\alpha^\circ F(X) R \subseteq X$
5	$\Leftrightarrow (S)^\circ (R) \subseteq X$	$\Rightarrow (R) \subseteq X$
		$X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$

$$([S_1, S_2]^\circ)^\circ ([R_1, R_2])^\circ = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



1	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$(R + S) [P, Q] = [R P, S Q]$
2	$= [S_1, S_2]^\circ [G(X) R_1, H(X) R_2]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
3	$= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$	

§6.4. Fast exponentiation and modulus computation

$\text{shift } f = (f \times 1) \ h \Rightarrow (\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f$ (6.4a)	
once f turns shift into h , the recursion that halves the argument at each step computes f	
(1)	(2)
(3)	(4)
1 $(\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h))$	$\text{convert}^\circ \llbracket [g, h] \rrbracket = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$
2 $= \text{convert}^\circ \llbracket [\text{zero } f, h] \rrbracket$	$\text{zero } f = g \wedge \text{shift } f = (f \times 1) \ h$
3 $= \text{convert}^\circ \text{ convert } f$	$\Rightarrow \text{convert } f = \llbracket [g, h] \rrbracket$
4 $\sqsubseteq f$	$\text{Simple}(\text{convert})$

$(\mu X : (\text{zero}^\circ \text{ one}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(a))) \sqsubseteq \text{exp}(a)$ (6.4b)	
the recursion that halves the exponent at each step computes a to the power b	
zero $\text{exp}(a) = \text{one}$	
a to the power 0 is 1	
shift $\text{exp}(a) = (\text{exp}(a) \times 1) \ \text{op}(a)$	
a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d	

$(\mu X : (\text{zero}^\circ \text{ zero}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(b))) \sqsubseteq \text{mod}(b)$ (6.4c)	
the recursion that halves a at each step computes the remainder of a by b	
zero $\text{mod}(b) = \text{zero}$	
$0 \text{ mod } b$ is 0	
shift $\text{mod}(b) = (\text{mod}(b) \times 1) \ \text{op}(b)$	
$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d	

§6.5. Unique fixed points

(6.5a)

$$R R \sqsubseteq S R \Rightarrow (X / R / S) R R \sqsubseteq X$$

((X/R)/S) followed by RR stays below X when $RR \sqsubseteq SR$

$$\text{inductive}(S) \wedge R R \sqsubseteq S R \Rightarrow \text{inductive}(R)$$

R is inductive when $RR \sqsubseteq SR$ for an inductive S

1	$(X / R / S) R R$	$R R \sqsubseteq S R$
2	$\sqsubseteq (X / R / S) S R$	$(R / S) S \sqsubseteq R$
3	$\sqsubseteq (X / R) R$	$(R / S) S \sqsubseteq R$
4	$\sqsubseteq X$	

§6.5.1. Membership

(6.5.1a)

member = $\emptyset$

a constant relator records no elements, so its membership is empty

member = [member, member]

a member of $F+G$ is a member of whichever summand the value is in

member = $(\pi_1 \text{ member}) \cup (\pi_2 \text{ member})$

a member of $F \times G$ is a member of either component

§6.5.2. Consequences

(6.5.2a)

$$\text{inductive}(S \text{ member}) \wedge X = S F(X) R \wedge Y = S F(Y) R \Rightarrow X = Y$$

when $S \text{ member}$ is inductive, two solutions X, Y of $X = SF(X)R$ are equal

(6.5.2b)

$$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S F(X) R \sqsubseteq X \Rightarrow \text{Entire}(X)$$

when $S \text{ member}$ is inductive and S, R are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire

(6.5.2c)

$$\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = g F(X) f \Rightarrow \text{Map}(X)$$

when $g \text{ member}$ is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

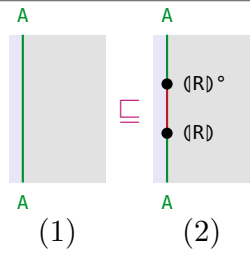
$$X = g F(X) f \wedge \text{Simple}(g) \wedge \text{Simple}(f) \wedge Y^\circ X \sqsubseteq 1 \Rightarrow (g F(Y) f)^\circ X \sqsubseteq 1$$

$(gF(Y)f)^\circ$ followed by X stays below 1 when g, f are simple and $Y^\circ X \sqsubseteq 1$

1	$(g F(Y) f)^\circ X$	$X = g F(X) f$
2	$= (g F(Y) f)^\circ g F(X) f$	$(R S)^\circ = S^\circ R^\circ$
3	$= f^\circ F(Y)^\circ g^\circ g F(X) f$	PreservesRecip(F)
4	$= f^\circ F(Y)^\circ g^\circ g F(X) f$	Simple(g)
5	$\sqsubseteq f^\circ F(Y)^\circ X f$	$Y^\circ X \sqsubseteq 1$
6	$\sqsubseteq f^\circ f$	Simple(f)
7	$\sqsubseteq 1$	

(6.5.2d)

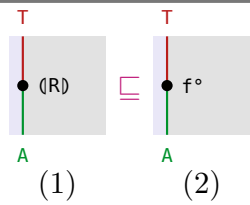
$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \implies 1 \sqsubseteq (R)^\circ (R)$
 when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective



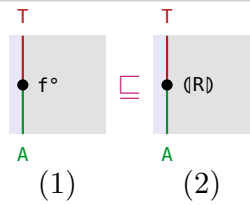
1	1	$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S$ $F(X) R \sqsubseteq X$ $\implies \text{Entire}(X) \text{ at } S=R^\circ$
2	$\sqsubseteq (R)^\circ (R)$	

(6.5.2e)

$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \implies f^\circ = (R)$
 a map f with $Rf \sqsubseteq F(f) \alpha$, for a surjective R , has the fold (R) as its converse



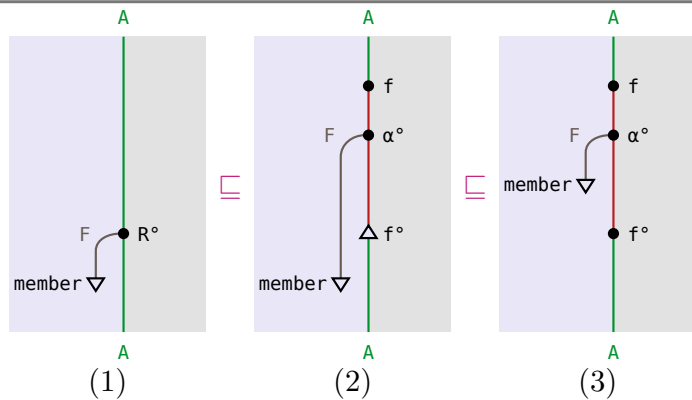
1	(R)	shunting f , fusion, $Rf \sqsubseteq F(f) \alpha$
2	$\sqsubseteq f^\circ$	



1	f°	$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R$ $\implies 1 \sqsubseteq (R)^\circ (R) \text{ and } (R)f \sqsubseteq 1$
2	$\sqsubseteq (R)$	

(6.5.2f)

$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \implies R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ$
 $R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



1	$R^\circ \text{ member}$	$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha$ $\implies R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$
2	$\sqsubseteq f \alpha^\circ F(f^\circ) \text{ member}$	$F(R) \text{ member} \sqsubseteq \text{member } R$
3	$\sqsubseteq f \alpha^\circ \text{ member } f^\circ$	

§6.6. Sorting by selection

§6.6.1. The function sort

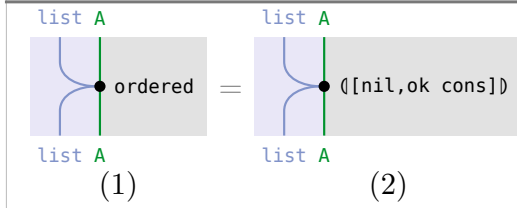
(6.6.1a)

definition	type	meaning
1 $\text{ordered}[] = \text{true}, \text{ordered}([a]\#xs) = (\forall x. x \in xs \Rightarrow R(a,x)) \wedge \text{ordered}(xs)$	$P[A]$	every element is R -below each element after it
2 $\text{ordered}[] = \text{true}, \text{ordered}([a]\#xs) = (\forall x. x \in xs \Rightarrow R(a,x)) \wedge \text{ordered}(xs)$	$[A] \rightarrow [A]$	the coreflexive that tests whether a list is ordered under R
3 $(a,xs) \text{ ok } q \iff (a,xs) = q \wedge (\forall x. x \in xs \Rightarrow R(a,x))$	$A \times [A] \rightarrow A \times [A]$	the coreflexive passing (a, x) when a is R -below every element of x

ordered = $\llbracket [\text{nil}, \text{ok cons}] \rrbracket$

(6.6.1b)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is R -below every element after it



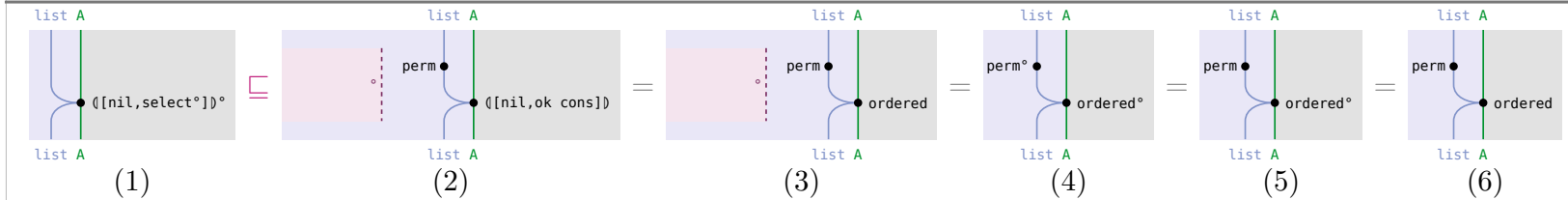
1 ordered	fold uniqueness
2 $= \llbracket [\text{nil}, \text{ok cons}] \rrbracket$	

§6.6.2. Selection sort

$\text{select}^\circ \sqsubseteq \text{ok cons perm} \implies \llbracket [\text{nil}, \text{select}^\circ] \rrbracket^\circ \sqsubseteq \text{perm ordered}$

(6.6.2a)

every output of unfolding the input by **select** is a sorted permutation of it

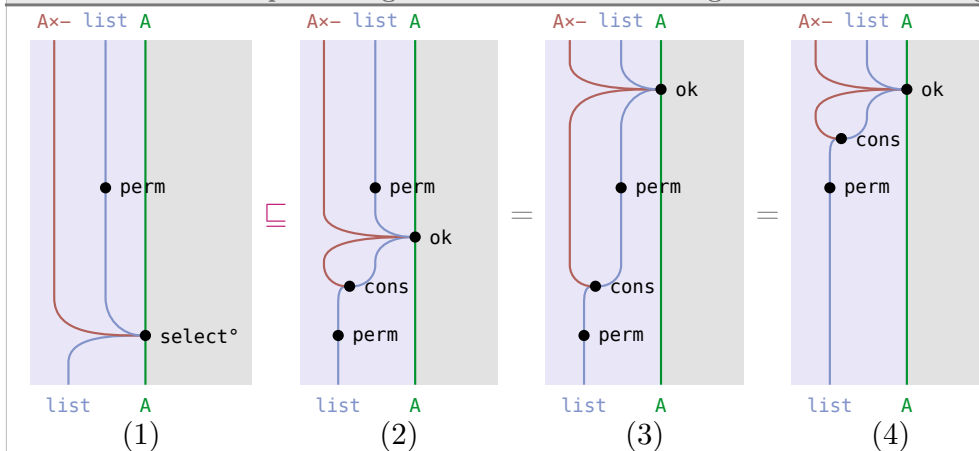


1 $\llbracket [\text{nil}, \text{select}^\circ] \rrbracket^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
2 $\sqsubseteq (\llbracket [\text{nil}, \text{ok cons}] \rrbracket \text{perm})^\circ$	$\implies \llbracket [\text{nil}, \text{select}^\circ] \rrbracket^\circ \sqsubseteq (\llbracket [\text{nil}, \text{ok cons}] \rrbracket \text{perm})^\circ$
3 $= (\text{ordered perm})^\circ$	(6.6.1b) $\text{ordered} = \llbracket [\text{nil}, \text{ok cons}] \rrbracket$
4 $= \text{perm}^\circ \text{ordered}^\circ$	$(R S)^\circ = S^\circ R^\circ$
5 $= \text{perm ordered}^\circ$	$\text{perm}^\circ = \text{perm}$
6 $= \text{perm ordered}$	$\text{ordered}^\circ = \text{ordered}$

$\text{select}^\circ \sqsubseteq \text{ok cons perm} \implies (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm}$

(6.6.2b)

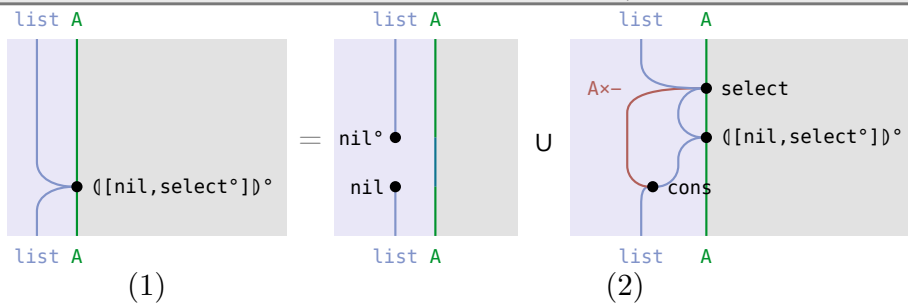
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



1 $(1 \times \text{perm}) \text{select}^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
2 $\sqsubseteq (1 \times \text{perm}) \text{ok cons perm}$	$\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$
3 $= \text{ok } (1 \times \text{perm}) \text{cons perm}$	$\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$
4 $= \text{ok cons perm}$	

$$([nil, select^\circ])^\circ = (nil^\circ nil) \cup (select (1 \times ([nil, select^\circ])^\circ) cons) \quad (6.6.2c)$$

selection sort returns $[]$ on $[]$, and otherwise selects (a, y) , sorts y and conses a back on



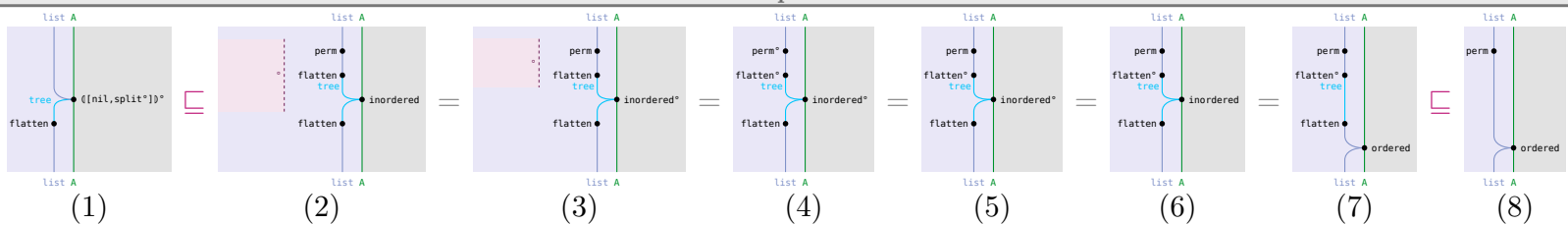
1	$[nil, select^\circ]^\circ$	unfolding the converse of a fold
2	$= (nil^\circ nil) \cup (select (1 \times ([nil, select^\circ])^\circ) cons)$	

§6.6.3. Quicksort

$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge split^\circ \sqsubseteq check' \text{ join perm} \Rightarrow ([nil, split^\circ])^\circ \quad (6.6.3a)$$

flatten $\sqsubseteq$ perm ordered

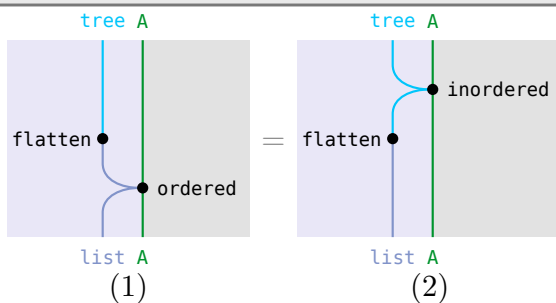
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



1	$[nil, split^\circ]^\circ \text{ flatten}$	$split^\circ \sqsubseteq check' \text{ join perm}$
2	$\sqsubseteq (inordered \text{ flatten perm})^\circ \text{ flatten}$	$\Rightarrow ([nil, split^\circ])^\circ \text{ flatten} \sqsubseteq (inordered \text{ flatten perm})^\circ \text{ flatten}$
3	$= (flatten \text{ perm})^\circ inordered^\circ \text{ flatten}$	$(R S)^\circ = S^\circ R^\circ$
4	$= perm^\circ flatten^\circ inordered^\circ \text{ flatten}$	$(R S)^\circ = S^\circ R^\circ$
5	$= perm \text{ flatten}^\circ inordered^\circ \text{ flatten}$	$perm^\circ = perm$
6	$= perm \text{ flatten}^\circ inordered \text{ flatten}$	$inordered^\circ = inordered$
7	$= perm \text{ flatten}^\circ \text{ flatten ordered}$	$(6.6.3b) (\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c))$
8	$\sqsubseteq perm \text{ ordered}$	$\Rightarrow \text{flatten ordered} = inordered \text{ flatten}$
		$flatten^\circ \text{ flatten} \sqsubseteq 1$

$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \Rightarrow \text{flatten ordered} = inordered \text{ flatten} \quad (6.6.3b)$$

flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening

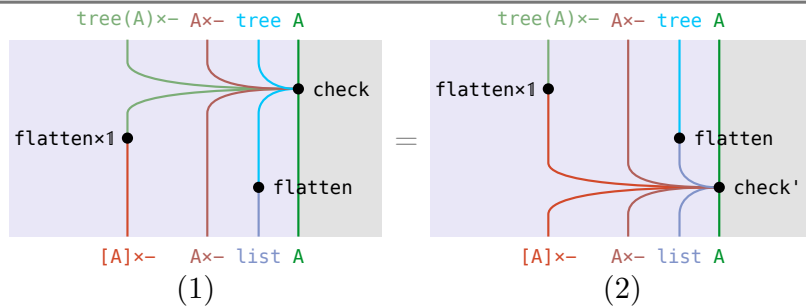


1	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
2	$= inordered \text{ flatten}$	

check (flatten × (1 × flatten)) = (flatten × (1 × flatten)) check'

(6.6.3c)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening

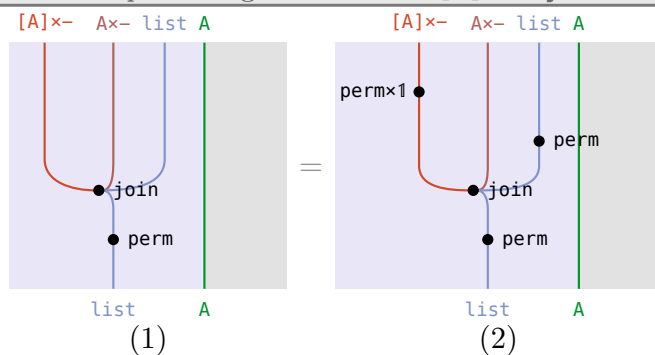


1	check (flatten × (1 × flatten))	b intree x ⇔ b inlist flatten(x)
2	= (flatten × (1 × flatten)) check'	

join perm = (perm × (1 × perm)) join perm

(6.6.3d)

permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list

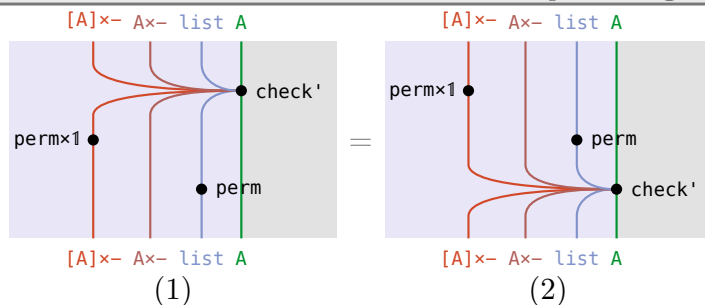


1	join perm	perm is transitive and closed under ++
2	= (perm × (1 × perm)) join perm	

check' (perm × (1 × perm)) = (perm × (1 × perm)) check'

(6.6.3e)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

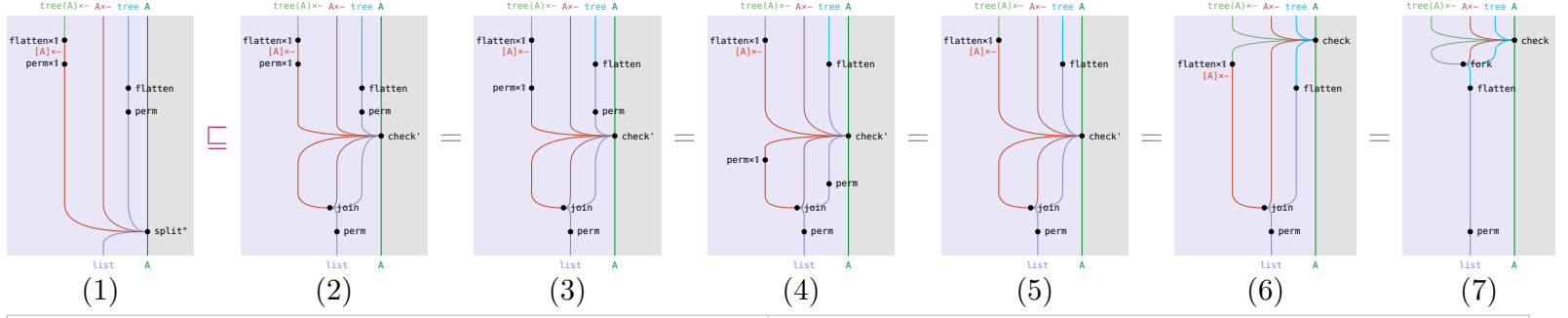


1	check' (perm × (1 × perm))	b inlist x ⇔ b inlist perm(x)
2	= (perm × (1 × perm)) check'	

$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ \sqsubseteq \text{check fork}$ (6.6.3f)

flatten perm

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted

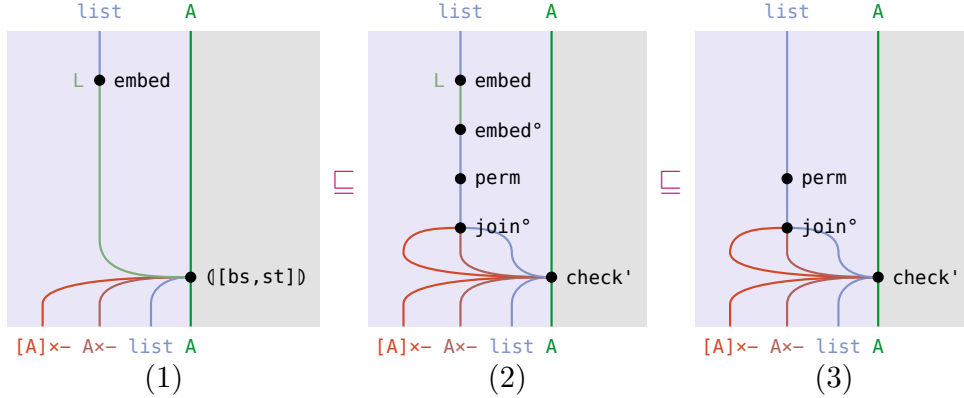


1	$((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$
2	$\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ check}' \text{ join perm}$	
3	$= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{ check}' \text{ join perm}$	(6.6.3e) $\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}'$
4	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$	(6.6.3d) $\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$
5	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \text{ join perm}$	(6.6.3c) $\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$
6	$= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{ join perm}$	$\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{ join}$
7	$= \text{check fork flatten perm}$	

$\text{bs} \sqsubseteq \text{wrap perm join}^\circ \text{ check}' \wedge (1 \times (\text{perm join}^\circ \text{ check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{ check}' \Rightarrow$ (6.6.3g)

embed $([\text{bs}, \text{st}]) \sqsubseteq \text{perm join}^\circ \text{ check}'$

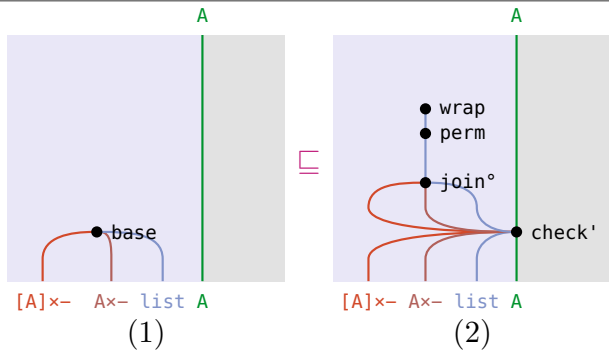
turning a list into a non-empty list and folding it with any $[\text{bs}, \text{st}]$ meeting the two conditions below gives only **checked** triples (x, a, y) whose $\text{join } x \# [a] \# y$ is a permutation of the list



1	$\text{embed } ([\text{bs}, \text{st}])$	$\text{bs} \sqsubseteq \text{wrap perm join}^\circ \text{ check}' \wedge (1 \times (\text{perm join}^\circ \text{ check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{ check}'$ $\Rightarrow \text{embed } ([\text{bs}, \text{st}]) \sqsubseteq \text{embed embed}^\circ \text{ perm join}^\circ \text{ check}'$
2	$\sqsubseteq \text{embed embed}^\circ \text{ perm join}^\circ \text{ check}'$	$\text{embed embed}^\circ \text{ perm join}^\circ \text{ check}' \sqsubseteq \text{perm join}^\circ \text{ check}'$
3	$\sqsubseteq \text{perm join}^\circ \text{ check}'$	

base $\sqsubseteq \text{wrap perm join}^\circ \text{ check}'$ (6.6.3h)

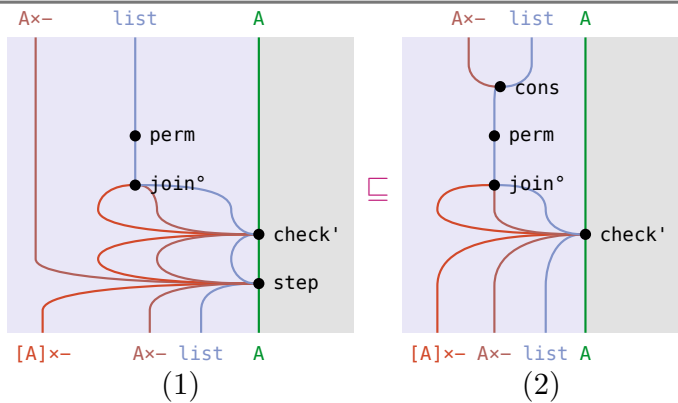
base(a) is a **checked** triple whose **join** is a permutation of the one-element list $[a]$



1	base	$[\] \# [a] \# [\] = [a]$, and check' holds on empty lists
2	$\sqsubseteq \text{wrap perm join}^\circ \text{ check}'$	

connected(R) $\Rightarrow$ (1 $\times$ (perm join $^\circ$ check')) step $\sqsubseteq$ cons perm join $^\circ$ check' (6.6.3i)

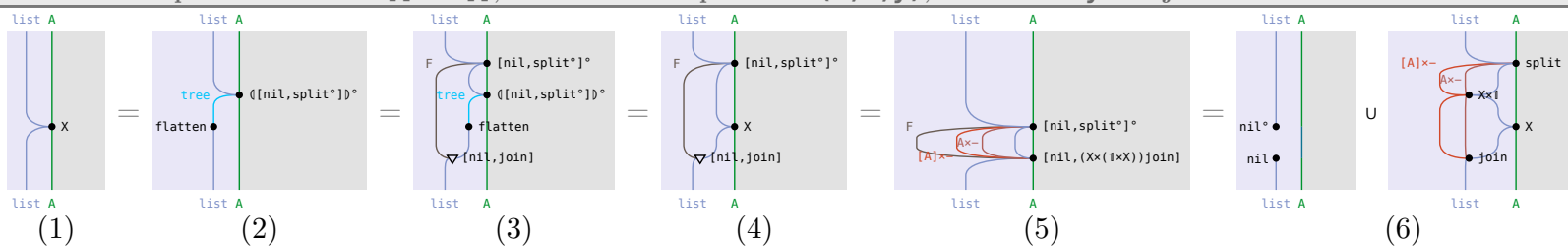
if (x,b,y) is a **checked** triple for a permutation of **l**, then **step(a, (x,b,y))** is a **checked** triple for a permutation of **[a] # l**



1	(1 $\times$ (perm join $^\circ$ check')) step	aRb puts a in front of x; otherwise bRa and a goes in front of y
2	$\sqsubseteq$ cons perm join $^\circ$ check'	

X = ([nil,split $^\circ$]) $^\circ$ flatten $\Rightarrow$ X = (nil $^\circ$ nil) u (split (X $\times$ (1 $\times$ X)) join) (6.6.3j)

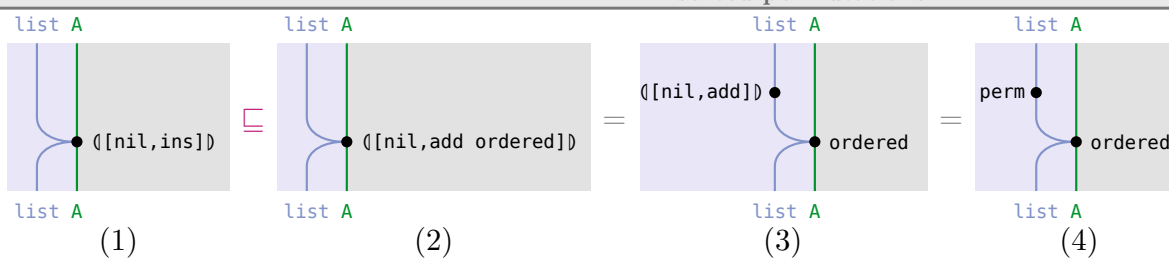
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



1	X	X = ([nil,split $^\circ$]) $^\circ$ flatten
2	= ([nil,split $^\circ$]) $^\circ$ flatten	S $^\circ$ F((S) $^\circ$ (R)) R = (S) $^\circ$ (R)
3	= [nil,split $^\circ$] $^\circ$ F([nil,split $^\circ$] $^\circ$ flatten) [nil,join]	X = ([nil,split $^\circ$]) $^\circ$ flatten
4	= [nil,split $^\circ$] $^\circ$ F(X) [nil,join]	F(S) [T,U] = [T,(S $\times$ (1 $\times$ S)) U]
5	= [nil,split $^\circ$] $^\circ$ [nil,(X $\times$ (1 $\times$ X)) join]	[U,V] $^\circ$ [R,S] = (U $^\circ$ R) u (V $^\circ$ S)
6	= (nil $^\circ$ nil) u (split (X $\times$ (1 $\times$ X)) join)	

(1 $\times$ ordered) ins $\sqsubseteq$ add ordered $\Rightarrow$ ([nil,ins]) $\sqsubseteq$ perm ordered (6.6.3k)

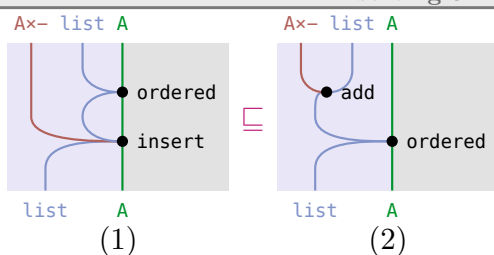
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



1	[nil,ins]	(1 $\times$ ordered) ins $\sqsubseteq$ add ordered
2	$\sqsubseteq$ [nil,add ordered]	$\Rightarrow$ ([nil,ins]) $\sqsubseteq$ ([nil,add ordered])
3	= [nil,add] ordered	[nil,add] ordered = [nil,add ordered]
4	= perm ordered	perm = [nil,add]

connected(R) $\wedge$ ($\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)$) $\Rightarrow$ (1 $\times$ ordered) insert $\sqsubseteq$ add ordered (6.6.3l)

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



1	(1 $\times$ ordered) insert	insert splices a in, and R transitive and connected
2	$\sqsubseteq$ add ordered	

§6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing R by composing it on the left and closing it by composing on the right give the same relation R^*

$$1 \in X \wedge X X \subseteq X \implies R \subseteq X \iff R^* \subseteq X \quad (6.7b)$$

X contains R exactly when it contains R^*

$$1 \in R^* \quad (6.7c)$$

R^* is reflexive

1	1	$R \subseteq R \cup S$
2	$\sqsubseteq 1 \cup (R R^*)$	$1 \cup (R R^*) = R^*$
3	$= R^*$	

$$R \subseteq R^* \quad (6.7d)$$

R^* contains R

1	R	
2	$= R$	$(6.7c) 1 \in R^*$
3	$\sqsubseteq R R^*$	$R R^* \subseteq R^*$
4	$\sqsubseteq R^*$	

$$(1 \cup (R (R^* / R^*))) R^* \subseteq R^* \quad (6.7e)$$

R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$, stated through division

$$R^* R^* \subseteq R^*$$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

1	$(1 \cup (R (R^* / R^*))) R^*$	$(S \cup T) R = (S R) \cup (T R)$
2	$= (R^*) \cup (R (R^* / R^*) R^*)$	
3	$= R^* \cup (R (R^* / R^*) R^*)$	$(R / R) R \subseteq R$
4	$\sqsubseteq R^* \cup (R R^*)$	$R R^* \subseteq R^*$
5	$\sqsubseteq R^*$	

$$1 \in X \wedge X X \subseteq X \wedge R \subseteq X \implies 1 \cup (R X) \subseteq X \quad (6.7f)$$

a reflexive transitive X containing R is a prefixed point of $Y \mapsto 1 \cup RY$

$$1 \in X \wedge X X \subseteq X \wedge R \subseteq X \implies R^* \subseteq X$$

a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

1	$1 \cup (R X)$	$1 \in X$
2	$\sqsubseteq X \cup (R X)$	$R \subseteq X$
3	$\sqsubseteq X \cup (X X)$	
4	$\sqsubseteq X$	

$$\frac{R^*}{\exists} = \left(\frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup} \quad (6.7g)$$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors

1	$\frac{R^*}{\exists}$	$1 \cup (R R^*) = R^*$
2	$= \frac{1 \cup (R R^*)}{\exists}$	$\frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{ cup}$
3	$= \left(\frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$	

§6.7.1. Computing closure

$R - \emptyset = R$	(6.7.1a)
taking nothing away leaves R	
$R \cup (S - R) = R \cup S$	
beside R , only the part of S outside R adds anything	
$R - (S \cup T) = R - S - T$	
taking away S and then T takes away $S \cup T$	
$(R \cup S) - T = (R - T) \cup (S - T)$	
taking T away from a union takes it away from each part	

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$	(6.7.1b)
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ	

$\mathbf{S} \mathbf{R}^* = (\mu \mathbf{X} : \mathbf{S} \cup (\mathbf{X} \mathbf{R}))$	(6.7.1c)
$\mathbf{S}$ followed by any number of $\mathbf{R}$ steps is the least $\mathbf{X}$ containing $\mathbf{S}$ and closed under a further $\mathbf{R}$ step	
$\mathbf{R}^* \mathbf{S} = (\mu \mathbf{X} : \mathbf{S} \cup (\mathbf{R} \mathbf{X}))$	
any number of $\mathbf{R}$ steps followed by $\mathbf{S}$ is the least $\mathbf{X}$ containing $\mathbf{S}$ and closed under an $\mathbf{R}$ step in front	

$\theta(\emptyset, S) = S R^*$ started with nothing found and S to explore, θ computes SR^*		(6.7.1d)
1	$\theta(\emptyset, S)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2	$\triangleq \emptyset \cup (\mu X : S \cup ((X R) - \emptyset))$	(6.7.1a) $R - \emptyset = R$
3	$= \emptyset \cup (\mu X : S \cup (X R))$	$\emptyset \cup R = R$
4	$= (\mu X : S \cup (X R))$	(6.7.1c) $S R^* = (\mu X : S \cup (X R))$
5	$= S R^*$	

$\theta(P, \emptyset) = P$		(6.7.1e)
with nothing left to explore, θ returns what it has found		
1	$\theta(P, \emptyset)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2	$\triangleq P \cup (\mu X : \emptyset \cup ((X R) - P))$	$(\mu X : \emptyset \cup ((X R) - P)) = \emptyset$
3	$= P \cup \emptyset$	$R \cup \emptyset = R$
4	$= P$	